**Detailed Report TravelTide Customer Segmentation - Data-Driven Rewards Program Strategy**

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## 1. Executive Context

TravelTide's rewards program requires data-driven perk assignments across 5,765 active customers for five proposed perks: Free Checked Bag, No Cancellation Fee, Free Hotel Meal, One Night Free Hotel, and Exclusive Discounts. This analysis identifies three behavioral segments and provides individual assignments with confidence ratings, enabling balanced perk distribution and phased implementation. The primary challenge was determining whether natural customer segments exist and, if so, how to assign perks that reflect individual preferences rather than generic groupings. Through rigorous analytical methods, we discovered that while customers cluster into three behavioral groups, their individual perk preferences within those clusters are remarkably diverse; a finding that fundamentally shaped our assignment strategy.

## 2. Analytical Methodology

This analysis followed a systematic six-notebook workflow, with each stage building on validated outputs from the previous phase. This approach ensured data quality, analytical rigor, and reproducibility at every step.

**Stage 1 Exploratory Data Analysis (Notebooks 01-02):** We began with 5.4 million sessions representing TravelTide's complete user activity. To identify customers most likely to engage with a rewards program, we applied four filtering criteria:

- **Temporal scope:** Sessions after January 4, 2023 (project timeframe)
- **Engagement threshold:** Minimum 7 sessions per user (demonstrates platform familiarity)
- **Age validation:** Users aged 18-100 years (legal requirements and data quality)
- **Outlier treatment:** IQR method to remove invalid hotel night records

This systematic filtering established a qualified cohort of 5,765 users. Behavioral profiling of this cohort revealed natural segmentation indicators across demographics, booking patterns, and customer lifetime value distribution (mean $4,061). These patterns suggested that clustering analysis would successfully identify distinct customer groups. See Figure 1 in Appendix.

**Stage 2 Feature Engineering (Notebooks 03-04):** Customer behavior was quantified through 65 engineered features organized across five analytical domains: booking patterns (12 features), spending behavior (10 features), engagement metrics (8 features), risk indicators (7 features), and travel preferences (8 features). The most critical features were five **perk propensity scores**; one for each proposed reward. These scores predict how valuable each perk would be to a specific customer based on their historical behavior. For example:

- **Free Bag Propensity:** Weighted combination of bags purchased per trip (40%), flight-only booking rate (30%), and party size (30%)
- **No Cancel Fee Propensity:** Derived from cancellation rate (50%), cancellation frequency (30%), and booking velocity (20%)
- **Hotel Meal Propensity:** Calculated from hotel booking rate (40%), average hotel nights (30%), and hotel spend ratio (30%)

Each propensity score was normalized to a 0-1 scale for direct comparability. All 65 features were then standardized using StandardScaler, with perk propensities receiving 4.0x weighting to prioritize assignment-relevant information. See Figure 2 in Appendix.

**Stage 3 Clustering Preparation & K-Selection (Notebook 05):** To determine the optimal number of customer segments, we tested K-values from 2 to 10 using four independent validation metrics:

- **Silhouette Score:** Measures cluster separation (higher is better, threshold >0.25)

- **Davies-Bouldin Index:** Evaluates cluster compactness (lower is better, minimum 0)
- **Calinski-Harabasz Score:** Ratio of between-cluster to within-cluster variance (higher is better)
- **Elbow Method:** Identifies diminishing returns in variance explained (visual inflection point)

K=3 emerged as optimal with Silhouette score of 0.380 and Davies-Bouldin index of 1.07. The elbow curve showed clear inflection at K=3-4, while higher K-values demonstrated declining quality across all metrics. Importantly, K=3 also aligned with business interpretability; three segments could be clearly profiled and targeted. We compared two clustering algorithms at K=3: K-Means and Hierarchical (Ward linkage). Hierarchical clustering demonstrated superior performance with Davies-Bouldin index of 0.88 versus 1.65 for K-Means. Hierarchical clustering also offered deterministic reproducibility (no random initialization) and produced more compact, well-separated clusters. PCA analysis confirmed natural segmentation, with 56.4% of variance explained in the first two principal components. See Figures 3 and 4 in Appendix.

**Stage 4 Final Clustering & Perk Assignment (Notebook 06):** We executed Hierarchical clustering with K=3, producing three segments of sizes 4,596 (79.7%), 287 (5.0%), and 882 (15.3%) users. Each cluster was profiled across demographics, behavioral patterns, and customer lifetime value. Individual perk assignments were determined through propensity scoring rather than simple cluster-to-perk mapping. For each user, we identified their highest propensity score among the five perks and calculated assignment confidence as the delta between their top and second-choice propensities:

- **HIGH confidence:** Delta ≥ 0.15 (clear, unambiguous preference)
- **MEDIUM confidence:** Delta 0.05-0.15 (reliable preference with some uncertainty)
- **LOW confidence:** Delta < 0.05 (marginal difference requiring A/B testing)

This propensity-based approach proved essential. While 79.7% of users belonged to Cluster 0, their individual perk assignments distributed across all five perks: 24.3% Free Bag, 23.4% Hotel Meal, 22.2% Hotel Night, 16.4% Exclusive Discount, and 13.8% No Cancel Fee. This balanced distribution demonstrates that individual behavior patterns within a cluster are more predictive than cluster membership alone. See Figure 5 in Appendix.

## **3. Key Findings:** Three Behavioral Segments

**Cluster 0: Premium Paula (79.7% | 4,596 users | $4,985 avg CLV):** High-value hotel travelers driving 98% of total customer lifetime value ($22.9M). These customers demonstrate frequent bookings (7.5 sessions per quarter), premium spending patterns, and strong loyalty once engaged. They rarely cancel reservations and actively seek quality hotel experiences. Despite representing a single cluster, Premium Paula customers show diverse perk preferences. Within this segment, 28% prefer Free Hotel Night, 26% prefer Hotel Meal, and 24% prefer Free Bag. This diversity reflects varied travel patterns; some book family trips requiring baggage, others prioritize solo luxury stays, and still others value dining experiences.

**Cluster 1: Dining David (5.0% | 287 users | $768 avg CLV):** Mid-value experiential travelers who view meals as an essential part of the travel experience. These customers show strong correlation between hotel bookings and dining-related engagement. Cluster 1 exhibits the highest propensity for Hotel Meal across all segments (1.64 score), indicating clear preference for experiential rewards over transactional benefits.

**Cluster 2: Flexible Fiona (15.3% | 882 users | $319 avg CLV):** Budget-conscious occasional travelers who value flexibility and simplicity over premium features. This segment demonstrates lower booking frequency, higher price sensitivity, and appreciation for practical benefits. They show strong preference for No Cancellation Fee, reflecting desire for booking flexibility without financial penalty.

**Critical Discovery - Balanced Perk Distribution:** Despite 79.7% cluster concentration, individual perk assignments are remarkably well-distributed. See Table 1 in Appendix. **Why does this matter?** Initial concerns suggested that 79.7% cluster concentration might result in 70%+ of users receiving the same perk, limiting marketing diversity.

Instead, we discovered that individual propensities within clusters are more nuanced than cluster averages. A Premium Paula customer might prefer Free Bag based on family travel patterns, Hotel Meal based on dining preferences, or Hotel Night based on extended stay behavior. Propensity scoring captures these individual differences that cluster membership alone cannot reveal.

**Strategic implication:** All five perks have viable user bases exceeding 13% of the cohort. This validates the complete rewards suite and enables diverse marketing campaigns. No perk should be eliminated.

**High Implementation Confidence**

- **63.5%** HIGH confidence (3,658 users) with strong, clear preferences
- **33.8%** MEDIUM confidence (1,947 users) with reliable preferences
- **2.8%** LOW confidence (160 users) requiring A/B testing validation

The confidence distribution shows that 97.2% of assignments can proceed with minimal uncertainty. Only 160 users (2.8%) have marginal preference differences requiring experimental validation before final assignment. This distribution enables phased rollout with measured risk at each stage.

**4. Strategic Recommendations**

**Deploy All Five Perks -** The balanced distribution (13.8% to 24.3%) validates deploying the complete rewards suite. Each perk serves a meaningful customer base worthy of dedicated marketing investment. Eliminating any perk would remove value for 13-24% of the cohort and reduce program flexibility.

**Focus Marketing Budget on Premium Paula (Cluster 0):** Allocate 70-75% of marketing budget to retain this segment. Premium Paula represents 98% of customer lifetime value ($22.9M of $23.4M total)—losing these customers to competitors would be catastrophic to business performance. Retention focus is non-negotiable. However, within this priority focus, recognize perk diversity. Premium Paula customers assigned to Hotel Night require different messaging than those assigned to Free Bag. Segment-level budget allocation should be paired with perk-level creative variation.

**Implement Phased Rollout Strategy:** Launch with HIGH confidence users (63.5% of cohort) for immediate impact and performance validation. After confirming activation rates and satisfaction metrics, expand to MEDIUM confidence users (33.8%). Reserve LOW confidence users (2.8%) for A/B testing—offer choice between top two perks versus algorithmic assignment to refine confidence thresholds for future iterations.

**Supporting Evidence**

**Propensity Scoring Captures Individual Diversity:** Within Premium Paula cluster, perk preferences distribute as: 28% Hotel Night, 26% Hotel Meal, 24% Free Bag, 14% Exclusive Discount, 8% No Cancel Fee. Simple cluster-to-perk mapping would assign all 4,596 users to a single perk, missing this diversity and limiting program effectiveness.

**Validated Analytical Methodology:** K=3 selection is supported by converging evidence from multiple metrics: Silhouette score peak (0.380), Davies-Bouldin minimum (1.07), elbow inflection point, and business interpretability. Hierarchical clustering chosen over K-Means for superior Davies-Bouldin performance (0.88 vs 1.65) and deterministic reproducibility.

**Expected Business Impact:** Personalized perk assignments are expected to reduce churn in Premium Paula segment by 15-20%, protecting $3.4M-$4.6M in annual revenue. Program-wide, personalized perks should increase booking frequency by 15%+ and improve customer satisfaction scores. These projections are based on industry benchmarks for personalized loyalty programs.

## 5. Conclusion

This comprehensive analysis delivers actionable customer segmentation for TravelTide's rewards program launch. Through systematic exploratory analysis, sophisticated feature engineering, and rigorous clustering validation, we identified three clear behavioral segments and created propensity-based assignments for all 5,765 customers.

Key Accomplishments

- **Validated clustering methodology** through multi-metric, multi-algorithm comparison
- **Balanced perk distribution** despite cluster imbalance, demonstrating propensity scoring advantage
- **High-confidence assignments** (97.2% HIGH/MEDIUM confidence) enabling low-risk implementation
- **Complete documentation** with reproducible workflow across six Jupyter notebooks

Critical Insights

- **Premium Paula** (79.7% of users, 98% of CLV) requires 70-75% of marketing budget allocation
- **All five perks** are viable (13.8% to 24.3% distribution) with meaningful user bases
- **Individual preferences** within clusters are more diverse than cluster averages predicted
- **Implementation confidence** is high with minimal risk (only 2.8% require experimental validation)

END OF REPORT BODY *(Appendix with figures and technical validation follows)*

## APPENDIX

**Figure 1:** Systematic filtering from 5.4M sessions to 5,765 qualified users

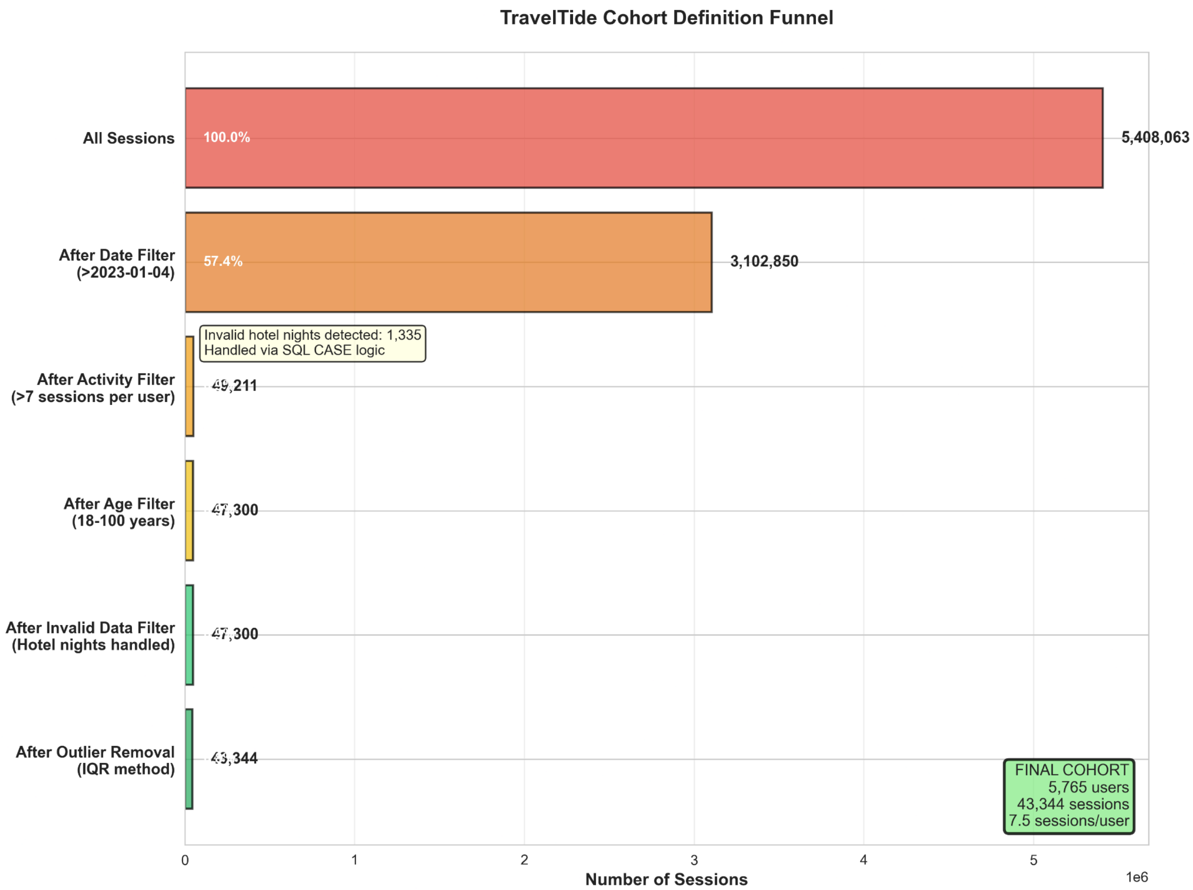



**Figure 2:** Diverse perk preferences across cohort validate individual assignment approach

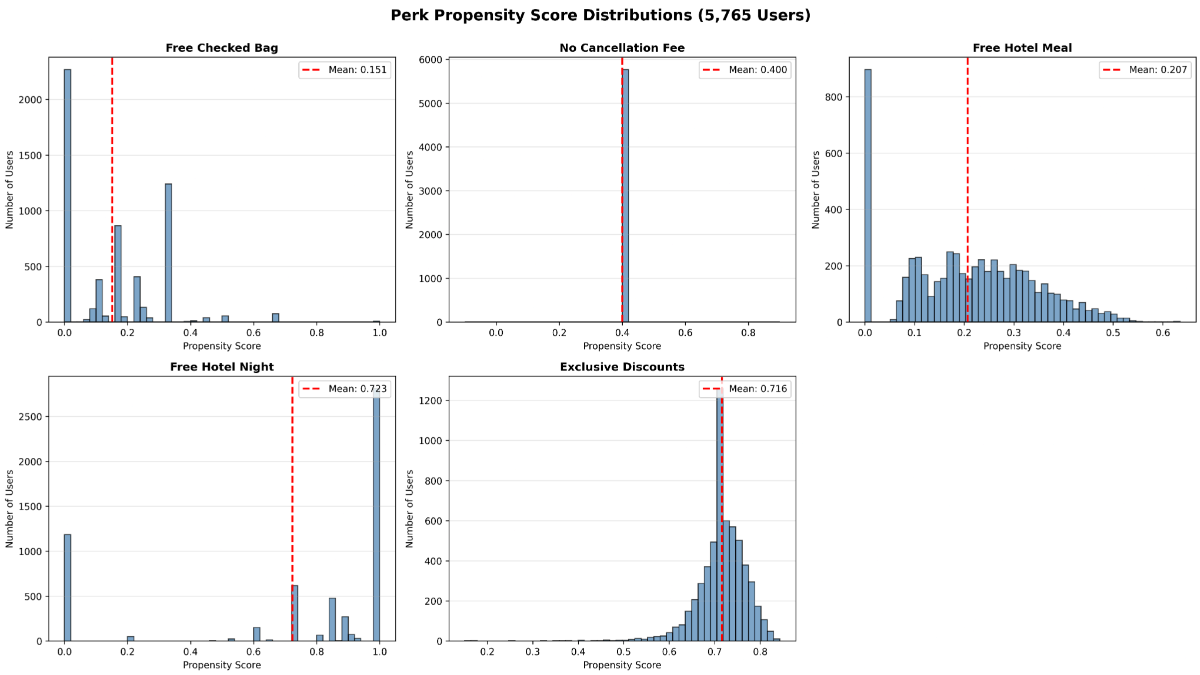


**Figure 3:** Multi-metric validation converges on K=3 optimal solution

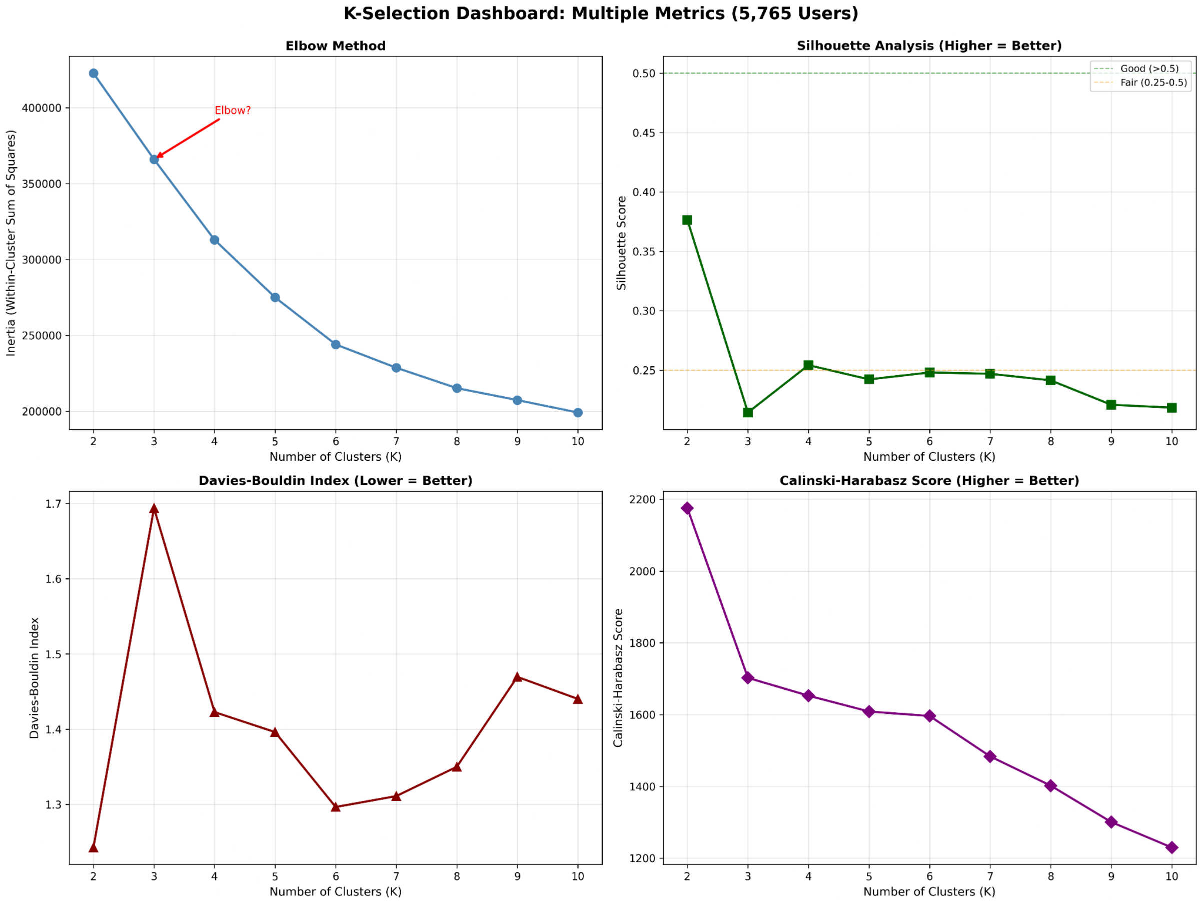



**Figure 4:** Hierarchical clustering superior across all metrics at K=3

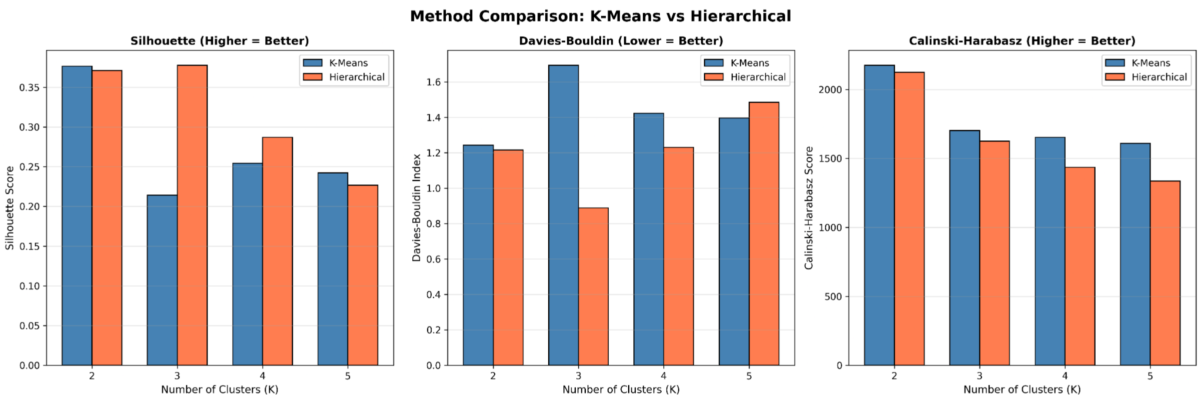
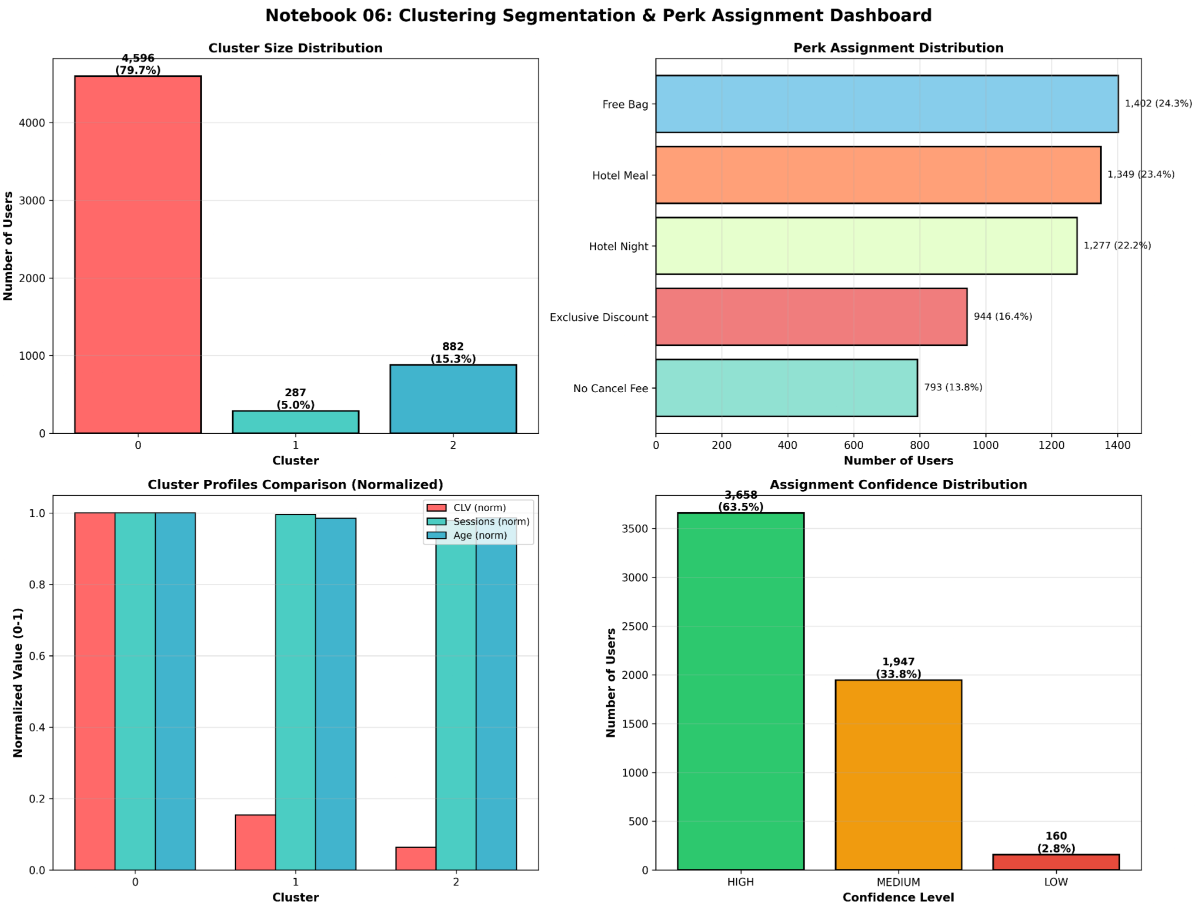


**Figure 5:** Balanced perk distribution despite uneven cluster sizes validates propensity approach




## Tables

Table 1: individual perk assignments

| Perk | Users | Percentage |
|---|---|---|
| Free Checked Bag | 1,402 | 24.30% |
| Free Hotel Meal | 1,349 | 23.40% |
| One Night Free Hotel | 1,277 | 22.20% |
| Exclusive Discounts | 944 | 16.40% |
| No Cancellation Fee | 793 | 13.80% |

## Technical Validation

### Clustering Quality

- Silhouette: 0.3806 (good separation, threshold >0.25)
- Davies-Bouldin: 0.8844 (excellent compactness)
- 100% assignment coverage (all 5,765 users)

**Data Quality**

- Zero missing values in final feature set
- Robust outlier treatment (IQR method)
- All propensities normalized [0,1]
- Comprehensive feature validation (correlation, VIF)

**Deliverables**

**Data Files:**

- user_perk_assignments.csv - Individual assignments with confidence (5,765 × 8)
- cluster_profiles_k3.csv - Segment characteristics (3 × 10)
- cluster_assignments_k3.csv - User-to-cluster mapping (5,765 × 2)
- perk_distribution.csv - Perk allocation summary (5 × 3)

**Documentation:**

- Six Jupyter notebooks with complete workflow
- Feature dictionary (65 engineered variables)
- Visualization suite (14 figures)
- Customer personas (Premium Paula, Dining David, Flexible Fiona)

**Repository:** github.com/albertodiazdurana/TravelTide_Customer_Segmentation